COMSATS University Islamabad

Registrar Office, Principal Seat, Islamabad

***

Scheme of Studies of Bachelor of Science in Computer Science for Fall 2026

1. Minimum Duration: **4 Years**

2. Minimum Semesters: **8**

2. Maximum Duration: **6 Years**

4. Maximum Semesters: **12**

Course Work	Min No. of Courses	Min No. of Credit Hrs
1 General Education	15	34
2 Computing Core Courses	14	48
3 Specialization Electives	8	24
4 Professional Certification		3
5 Interdisciplinary/Allied Courses	4	12
6 Field Experience / Internship		3
7 Capstone Project — 6 credit hours		6

Note:

- **Students are required to complete their academic programs in strict adherence to the applicable and officially notified Schemes of Studies (SoS). All updated rules, regulations, and policies, as periodically amended, shall be enforced.**
- **Prior to course registration, the Campus Registration Office must conduct a thorough consolidation and comparison between the hard copies of the signed notifications of the applicalbe Scheme of Studies and the versions uploaded on CU Online. This verification process must be completed before courses are made available for student registration, ensuring the elimination of any discrepancies or errors in the SoS or during the registration process.Should any discrepancies or mismatches be identified between the hard copy of the signed notifications and the SoS on CU Online, it must be got corrected from Registration Office, PS on the CU Online SoS to align with the officially signed hard copy prior to course offering.**
- **This procedure is designed to address and rectify potential issues at the course registration stage, thereby preventing complications during the awarding of degrees or the issuance of Final Result Notifications(FRNs) at CUI System.In the event that any course is registered with incorrect details, the responsibility will lie with the concerned campus course registration office, and appropriate corrective actions will be enforced.**

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5 Capstone project

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
1	CSC498	Final Year Project I	2(0, 2)		
2	CSC499	Final Year Project II	4(0, 4)		

a. General Education Courses:

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
3	CSC101	Applications of Information and Communication Technologies	3(2, 1)		
4	HUM104	Functional English	3(3, 0)		
5	HUM112	Islamic Studies	2(2, 0)		
6	HUM113	Ideology and Constitution of Pakistan	2(2, 0)		
7	HUM116	Ethics	2(2, 0)		
8	HUM117	Pakistan Studies	2(2, 0)		
9	HUM120	Expository Writing	3(3, 0)		
10	HUM161	Understanding of Holy Quran - I (Fehme-Quran-I)	1(0, 1)		
11	HUM162	Understanding of Holy Quran - II (Fehme-Quran-II)	1(0, 1)		
12	HUM208	Civics and Community Engagement	2(2, 0)		
13	MGT250	Introduction to Entrepreneurship	2(2, 0)		
14	MTH103	Exploring Quantitative Skills	3(3, 0)		
15	MTH114	Tools for Quantitative Reasoning	3(3, 0)		
16	XXXXXX	Art & Humanities (one course)	2(2, 0)		
17	XXXXXX	Natural Sciences (one course)	3(2, 1)		
18	XXXXXX	Social Sciences (one course)	2(2, 0)		

Art & Humanities Elective (choose one)

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
19	ADA111	History and Theory of Art and Culture I	2(2, 0)		
20	ADA111	History and Theory of Art and Culture I	2(2, 0)		
21	ARC108	Art Appreciation	2(2, 0)		
22	HUM310	Islamic History	2(2, 0)		

Capstone Project: /

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
23	CSC498	Final Year Project I	2(0, 2)		CSC241 ,CSC270 ,CSC291
24	CSC499	Final Year Project II	4(0, 4)		CSC498

Computer Science Core Courses (14 courses to complete 48 credit hours)

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
25	CSC103	Introduction to Computers and Programming	4(3, 1)		
26	CSC211	Data Structures	4(3, 1)		CSC103
27	CSC232	Information Security	3(2, 1)		
28	CSC241	Object Oriented Programming	4(3, 1)		CSC103
29	CSC262	Artificial Intelligence	3(2, 1)		
30	CSC270	Database Systems	4(3, 1)		
31	CSC275	Computer Networks	3(2, 1)		
32	CSC291	Software Engineering	3(3, 0)		
33	CSC301	Design and Analysis of Algorithms	3(3, 0)		CSC211
34	CSC312	Theory of Automata	3(3, 0)		
35	CSC325	Computer Organization & Assembly Language	4(3, 1)		EEE241
36	EEE241	Digital Logic Design	4(3, 1)		

Computing Core Courses

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)

37	CSC103	Programming Fundamentals	4(3, 1)		
38	CSC232	Information Security	3(3, 0)		
39	CSC241	Object Oriented Programming	4(3, 1)		
40	CSC262	Artificial Intelligence	3(2, 1)		
41	CSC270	Database Systems	4(3, 1)		
42	CSC275	Computer Networks	3(2, 1)		
43	CSC291	Software Engineering	3(3, 0)		
44	CSC301	Design and Analysis of Algorithms	3(3, 0)		
45	CSC312	Theory of Automata	3(3, 0)		
46	CSC319	Cloud Computing	3(2, 1)		
47	CSC324	Operating System Principles	4(3, 1)		
48	CSC325	Computer Organization and Assembly Language	3(2, 1)		
49	EEE241	Digital Logic Design	4(3, 1)		

Deficiency Courses (only for Pre-Medical Students)

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
50	MTH091	Pre-Calculus I	3(3, 0)		

Deficiency Courses (only for Pre-Medical Students) /

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
51	MTH092	Pre-Calculus II	3(3, 0)		

Field Experience / Internship

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
52	CSC395	Internship	3(0, 3)		

General Education Courses: :

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
53	HUM141	Seerah of Holy Prophet S.A.W.	0(0, 0)		

Interdisciplinary/Allied Courses

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
54	CSC102	Discrete Structures	3(3, 0)		
55	CSC410	Professional Practices for IT	3(3, 0)		
56	DEV342	Local Government System and Rural Development	3(3, 0)		
57	ECO212	Economics of Pakistan	3(3, 0)		
58	ECO484	Project Planning and Monitoring	3(3, 0)		
59	FIN485	Technology in Finance	3(3, 0)		
60	HUM220	Introduction to Psychology	3(3, 0)		
61	HUM221	International Relations	3(3, 0)		
62	HUM430	French	3(3, 0)		
63	HUM431	German	3(3, 0)		
64	HUM432	Arabic	3(3, 0)		
65	HUM434	Chinese	3(3, 0)		
66	MET205	Geographic Information System	3(3, 0)		
67	MET317	Geospatial Databases	3(3, 0)		
68	MET411	Spatial Data Analysis and Modeling	3(2, 1)		
69	MGT131	Financial Accounting	3(3, 0)		
70	MGT210	Fundamentals of Marketing	3(3, 0)		
71	MGT350	Human Resource Management	3(3, 0)		
72	MGT513	New Product Development	3(3, 0)		
73	MTH104	Calculus and Analytic Geometry (Mandatory)	3(3, 0)		
74	MTH231	Linear Algebra (Mandatory)	3(3, 0)		
75	MTH240	Introduction to Scientific Computing	3(2, 1)		
76	MTH262	Statistics and Probability Theory	3(3, 0)		

List of Art & Humanities Courses (any one course from the following list)

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
77	HUM107	21st Century Communication Skills	2(2, 0)		

78	HUM123	Fundamentals of Philosophy	2(2, 0)		

Mandatory Courses

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
79	CSC203	Artificial Intelligence Essentials	3(3, 0)		

Natural Sciences Elective (choose one)

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
80	BIO100	Fundamentals of Biology	3(2, 1)		
81	BIO310	Introduction to Bioinformatics	3(2, 1)		
82	BIO321	Fundamentals of Genetics	3(2, 1)		
83	CHM100	Introduction to Chemistry	3(2, 1)		
84	PHY124	Applied Physics	3(2, 1)		

Professional Certification

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
85	CSC375	Professional Certification	3(0, 3)		

Social Sciences Elective (choose one)

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
86	HUM122	Fundamentals of Psychology	2(2, 0)		
87	HUM130	Fundamentals of Sociology	2(2, 0)		
88	HUM131	Anthropology	2(2, 0)		

89	HUM209	Fundamentals of Political Science	2(2, 0)		
90	HUM222	Fundamentals of International Relations	2(2, 0)		

Specialization Elective - Artificial Intelligence

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
91	AIC270	Programming for Artificial Intelligence	3(2, 1)		CSC262
92	AIC329	Multi-agent Systems	3(3, 0)		
93	AIC337	Agentic AI	3(2, 1)		
94	AIC341	Introduction to Computer Vision	3(2, 1)		
95	AIC345	Generative AI	3(3, 0)		
96	AIC350	Topics in Artificial Intelligence I	3(3, 0)		
97	AIC354	Machine Learning Fundamentals	3(2, 1)		
98	AIC355	Machine Learning Operations (MLOps)	3(2, 1)		
99	AIC357	Responsible AI	3(3, 0)		
100	AIC365	Natural Language Processing	3(2, 1)		
101	AIC368	Optimization Techniques	3(3, 0)		
102	AIC369	Stochastic Processes	3(3, 0)		
103	AIC372	Knowledge Representation and Reasoning	3(3, 0)		CSC262
104	AIC378	Speech Processing	3(2, 1)		
105	AIC380	Artificial Neural Networks and Deep Learning	3(2, 1)		CSC262
106	AIC456	Reinforcement Learning	3(2, 1)		
107	AIC469	Evolutionary Computing	3(3, 0)		
108	AIC483	Topics in Artificial Intelligence II	3(3, 0)		
109	DSC306	Data Mining	3(2, 1)		

Specialization Elective - Cyber Security

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
110	CYC205	Introduction to Cyber Security	3(2, 1)		
111	CYC207	Information Security Management	3(3, 0)		
112	CYC210	Digital Forensics and Incident Response	3(2, 1)		
113	CYC215	Data Center Security	3(3, 0)		
114	CYC230	Mathematical Foundations for Cryptography	3(3, 0)		
115	CYC232	Symmetric Key Cryptography	3(3, 0)		
116	CYC312	Cyber Threat Intelligence	3(3, 0)		
117	CYC315	Computer Forensics	3(3, 0)		
118	CYC317	Smart Devices and IoT Forensics	3(3, 0)		
119	CYC318	Information Privacy and Security	3(3, 0)		
120	CYC320	Asymmetric Key Cryptography	3(3, 0)		

121	CYC331	Cryptography	3(3, 0)		
122	CYC340	Cryptanalysis and Attack Techniques	3(3, 0)		
123	CYC350	Topics in Cyber Security I	3(3, 0)		
124	CYC365	Network Security	3(2, 1)		CSC275
125	CYC370	Machine Learning for Network Security	3(3, 0)		
126	CYC372	Network and System Administration	3(3, 0)		
127	CYC373	Network Forensics	3(3, 0)		
128	CYC386	Secure Software Design and Development	3(2, 1)		
129	CYC395	Vulnerability Assessment and Penetration Testing	3(2, 1)		
130	CYC400	Cryptographic Hash Functions and Blockchain Applications	3(3, 0)		
131	CYC403	Cloud and Application Security	3(2, 1)		
132	CYC405	Cloud Security	3(3, 0)		
133	CYC407	Cloud Forensics and Log Analysis	3(3, 0)		
134	CYC410	Open Source Intelligence Techniques	3(3, 0)		
135	CYC412	Incident Response Orchestration and Automation	3(3, 0)		
136	CYC460	Malware Analysis and Reverse Engineering	3(2, 1)		
137	CYC465	Security Automation for Network Operations	3(3, 0)		
138	CYC467	Mobile and Wireless Network Security	3(3, 0)		
139	CYC483	Topics in Cyber Security II	3(3, 0)		

Specialization Elective - Data Science

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
140	AIC341	Introduction to Computer Vision	3(2, 1)		
141	AIC345	Generative AI	3(3, 0)		
142	AIC354	Machine Learning Fundamentals	3(2, 1)		
143	AIC365	Natural Language Processing	3(2, 1)		
144	AIC380	Artificial Neural Networks and Deep Learning	3(2, 1)		CSC262
145	DSC293	Data Science Fundamentals	3(2, 1)		
146	DSC295	Data Engineering	3(2, 1)		
147	DSC300	Statistical Methods in Data Science	3(3, 0)		MTH262
148	DSC303	Tools and Techniques in Data Science	3(2, 1)		
149	DSC306	Data Mining	3(2, 1)		
150	DSC309	Data Science Design Patterns	3(3, 0)		
151	DSC327	Data Visualization Techniques	3(2, 1)		
152	DSC350	Topics in Data Science I	3(3, 0)		
153	DSC354	Data Warehousing and Business Intelligence	3(2, 1)		CSC270
154	DSC403	Social Media Mining	3(3, 0)		
155	DSC412	Text Mining	3(3, 0)		
156	DSC415	Big Data Analytics	3(2, 1)		DSC306
157	DSC483	Topics in Data Science II	3(3, 0)		

Specialization Elective - General Electives

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
158	CSC303	Mobile Application Development	3(2, 1)		CSC241
159	CSC316	Advance Database Systems	3(2, 1)		CSC270
160	CSC331	Digital Image Processing	3(2, 1)		
161	CSC336	Web Technologies	3(2, 1)		CSC241
162	CSC337	Advanced Web Technologies	3(2, 1)		CSC336
163	CSC353	Computer Graphics	3(2, 1)		
164	CSC381	Blockchain Technology	3(3, 0)		
165	CSC412	Advanced Programming	3(2, 1)		CSC241
166	CSC417	E-Commerce and Digital Marketing	3(2, 1)		CSC241
167	CSC418	DevOPs for Cloud Computing	3(2, 1)		
168	CSC441	Compiler Construction	3(2, 1)		CSC312
169	CSC454	Pattern Recognition	3(3, 0)		
170	CSC475	Numerical Computing	3(3, 0)		

Specialization Elective - Software Engineering

	Course Code	Course Title	Credit Hours	Co-requisite(s)	Prerequisite (s)
171	AIC354	Machine Learning Fundamentals	3(2, 1)		
172	CSC334	Parallel and Distributed Computing	3(2, 1)		CSC324
173	CSE302	Software Quality Engineering	3(2, 1)		CSC291
174	CSE303	Software Design and Architecture	3(3, 0)		CSC291
175	CSE305	Software Requirement Engineering	3(3, 0)		CSC291
176	CSE307	DevOps Principles and Practices	3(2, 1)		
177	CSE325	Software Construction and Development	3(2, 1)		CSE303
178	CSE326	Human Computer Interaction	3(3, 0)		
179	CSE327	Software Re-Engineering	3(3, 0)		CSC291
180	CSE337	Cross-platform Application Development	3(2, 1)		
181	CSE344	Green Software Engineering	3(3, 0)		
182	CSE350	Topics in Software Engineering I	3(3, 0)		
183	CSE356	Formal Methods	3(3, 0)		CSC291
184	CSE482	Automated Software Testing	3(2, 1)		
185	CSE483	Topics in Software Engineering II	3(3, 0)		
186	CSE494	Software Project Management	3(2, 1)		CSC291

Note:

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